Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **secure REST APIs, cloud integrations, SQL-backed services, and observable distributed systems**, with focused project experience in FHIR interoperability, Apigee policies, Azure messaging, and healthcare workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Advanced NLP, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Structures, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Architected FastAPI services and Azure SQL workflows for 500+ users, reducing dashboard latency 71% through **normalized schemas, indexed queries, row-level security, structured logging, monitoring, and systematic troubleshooting**.
- Processed 1K+ concurrent jobs at 3x throughput through **asynchronous orchestration, rate-aware queues, fault isolation, HTTP error handling, automated tests, deployment controls, and secure, reliable enterprise-wide system-to-system integrations**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Delivered React and Flask workflows supporting 150+ instructors and 10K+ daily requests through **REST APIs, authentication, payload validation, role-aware access, integration testing, production debugging, and responsive interfaces**.
- Built concurrent Python ingestion for PDF, DOCX, ZIP, OCR, and multimodal routing, reducing turnaround 60% while improving quality through **structured diagnostics, exception handling, edge-case tests, and failure recovery**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Designed REST integrations and cloud data pipelines using object-oriented patterns, AWS Glue, S3, and SQL, improving response times 30% while reducing technical debt 20% through **modular service boundaries**.
- Automated 20+ unit and integration suites across Windows and Linux using PyTest, Mockito, Jenkins, and virtual machines, strengthening **API quality, CI/CD validation, regression detection, and dependable releases**.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built React and Node.js microservices for resources, billing, permissions, and notifications, synchronizing 2M+ records through **REST APIs, WebSockets, JWT authentication, SQL-backed transactional workflows, and resilient service integrations**.
- Automated Docker and Jenkins deployments, reducing release time 50% while using Sentry, Azure Monitor, and Kibana for **APM, log aggregation, alerting, root-cause analysis, and continuous reliability improvement**.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Translated client requirements into technical designs for 12+ integrations, mapping **HTTP contracts, authentication flows, data synchronization, error responses, deployment dependencies, and implementation decisions** while reducing design errors 25%.
- Built Flask and Firebase APIs supporting 50K+ records with sub-200-millisecond responses, delivering **Postman validation, SQL analysis, integration debugging, technical documentation, stakeholder demonstrations, and responsive production support** through release.

Skills

Healthcare Interoperability: FHIR R4, HL7, Patient, Encounter, Observation, Bundle, Epic Integration, Healthcare APIs, Clinical Data Validation
API Management & Security: Apigee, API Proxies, Rate Limiting, Spike Arrest, Quotas, Fault Rules, OAuth 2.0, JWT, API Keys, Basic Authentication, HTTP Status Codes
Development & Testing: Python, Java, TypeScript, JavaScript, Node.js, FastAPI, Flask, REST, Postman, Bruno, SoapUI, Insomnia, OpenAPI, Integration Testing
Azure & Messaging: Azure Functions, Logic Apps, App Service, Containers, Cosmos DB, Event Hubs, Kafka, AMQP, WebSockets, CI/CD, Docker, Git
Data & Observability: SQL, PostgreSQL, Azure SQL, MySQL, NoSQL, Data Modeling, Datadog, APM, Log Aggregation, Dashboards, Real-Time Alerting, Debugging, Documentation

Projects

FHIR Gateway Exchange | *FHIR R4, Apigee, OAuth 2.0, JWT, Azure Functions, Cosmos DB*

- Designed a **secure FHIR API gateway** exposing Patient, Encounter, Observation, and Bundle resources with OAuth scopes, JWT validation, rate limits, spike arrest, quotas, fault flows, and auditable consent enforcement.

Clinical Integration Operations Hub | *Azure Event Hubs, Kafka, SQL, Datadog, Postman*

- Built an **observable healthcare integration pipeline** validating FHIR payloads, routing events, correlating HTTP failures, replaying dead letters, reconciling SQL records, and surfacing Datadog dashboards, alerts, traces, and actionable diagnostics.

Chat2SQL | *Java, TypeScript, SQL, Docker, Maven, Node.js* | Code

- Built a cross-platform **natural-language SQL workspace** supporting 10+ databases, schema synchronization, query validation, and Docker-based environments, enabling users to explore complex relational data through reproducible, database-aware workflows.